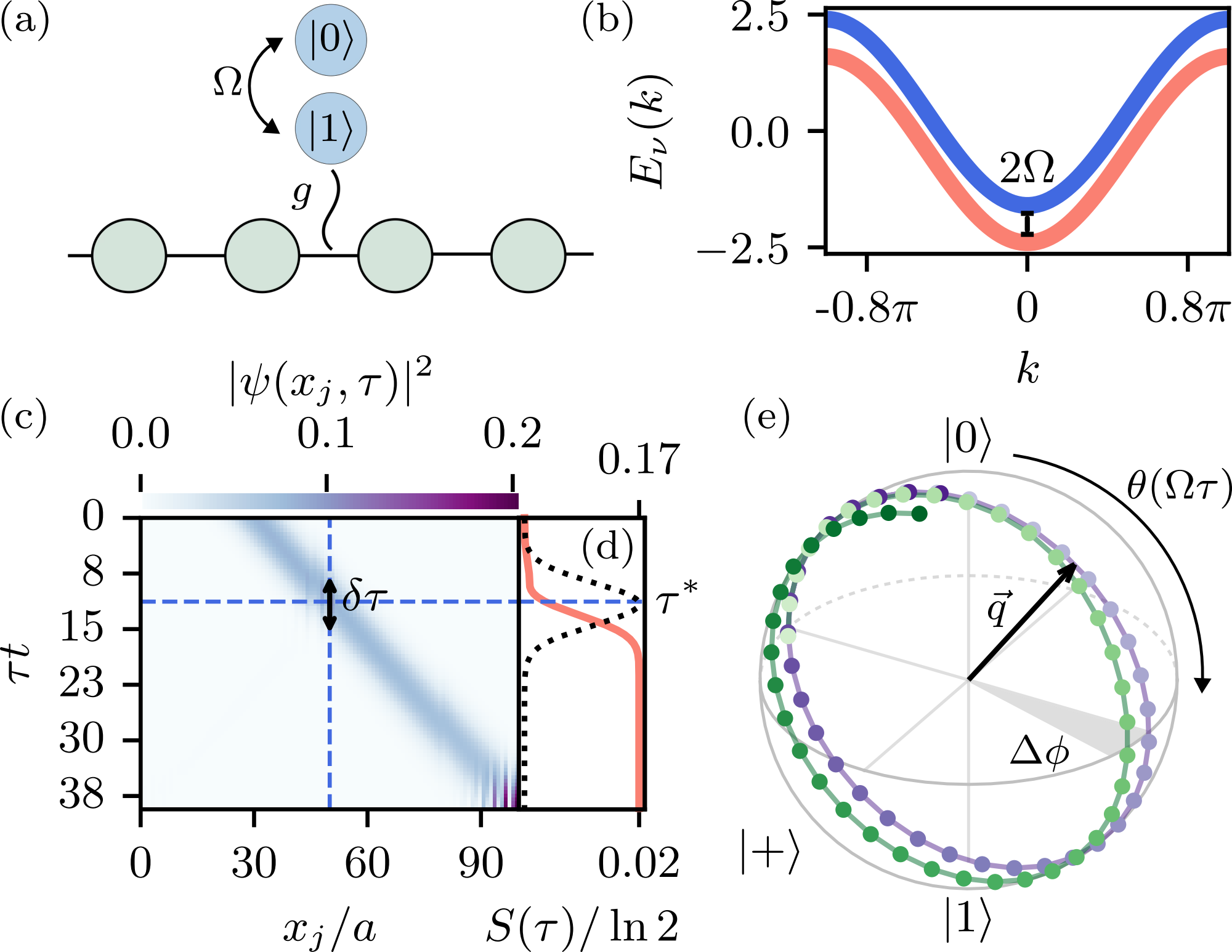

(a)
$|0\rangle$
Ω
$|1\rangle$
g
(b)
$E_\nu(k)$
2.5
0.0
−2.5
2Ω
-0.8π
0
0.8π
k
$|\psi(x_j,\tau)|^2$
(c)
0.0
0.1
0.2
0.17
0
8
15
23
30
38
τt
$\delta\tau$
(d)
τ^*
0
30
60
90
0.02
x_j/a
$S(\tau)/\ln 2$
(e)
$|0\rangle$
$\theta(\Omega\tau)$
$\vec{q}$
$\Delta\phi$
$|+\rangle$
$|1\rangle$